Example 1 Find three equivalent fractions of $\frac{3}{4}$ and $\frac{2}{7}$.

Solution

Equivalent Fraction of $\frac{3}{4}$	Equivalent Fraction of $\frac{2}{7}$
$\frac{3}{4}=\frac{3\times2}{4\times2}=\frac{6}{8}$	$\frac{2}{7}=\frac{2\times2}{7\times2}=\frac{4}{14}$
$\frac{3}{4}=\frac{3\times3}{4\times3}=\frac{9}{12}$	$\frac{2}{7}=\frac{2\times3}{7\times3}=\frac{6}{21}$
$\frac{3}{4}=\frac{3\times4}{4\times4}=\frac{12}{16}$	$\frac{2}{7}=\frac{2\times4}{7\times4}=\frac{8}{28}$

Equivalent fractions of $\frac{3}{4}$: $\frac{3}{4}=\frac{6}{8}=\frac{9}{12}=\frac{12}{16}$

Equivalent fractions of $\frac{2}{7}$: $\frac{2}{7}=\frac{4}{14}=\frac{6}{21}=\frac{8}{28}$

Try these

Find the unknown in the following equivalent fractions

i) $\frac{3}{5}=\frac{9}{\square}$ ii) $\frac{\square}{7}=\frac{16}{28}$ iii) $\frac{\square}{3}=\frac{10}{15}$ iv) $\frac{42}{48}=\frac{\square}{8}$

1.1 Introduction

Fractions are used in life situations such as

- To express time as quarter past 3, half past 4, quarter to 5.
- To say the quantum of work completed as quarter / half / three quarters of the work completed.
- To say the distance between two places as half a kilometre / two and half kilometre.
- To express the quantity of ingredients to be used in a recipe as half of the rice taken, half of the dhal taken etc.

Mathematics Alive - Fractions in Real Life



Nine-Tenths of water on the earth is salty.



The distance between Chennai Egmore and the Directorate of Public Instruction Campus (DPI) is nearly $1\frac{1}{2}$ kilometre.

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